

Lab Report: 8

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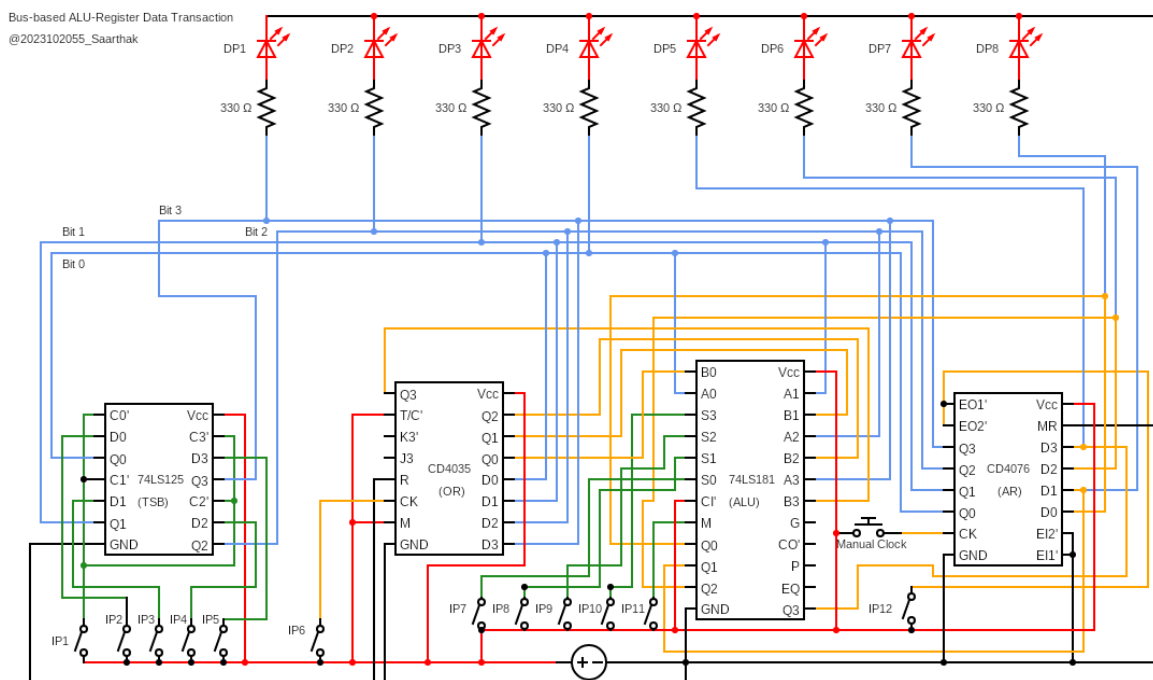
Group number: 6

Objective: To implement a bus-based ALU-Register data transaction system

Electronic components used:

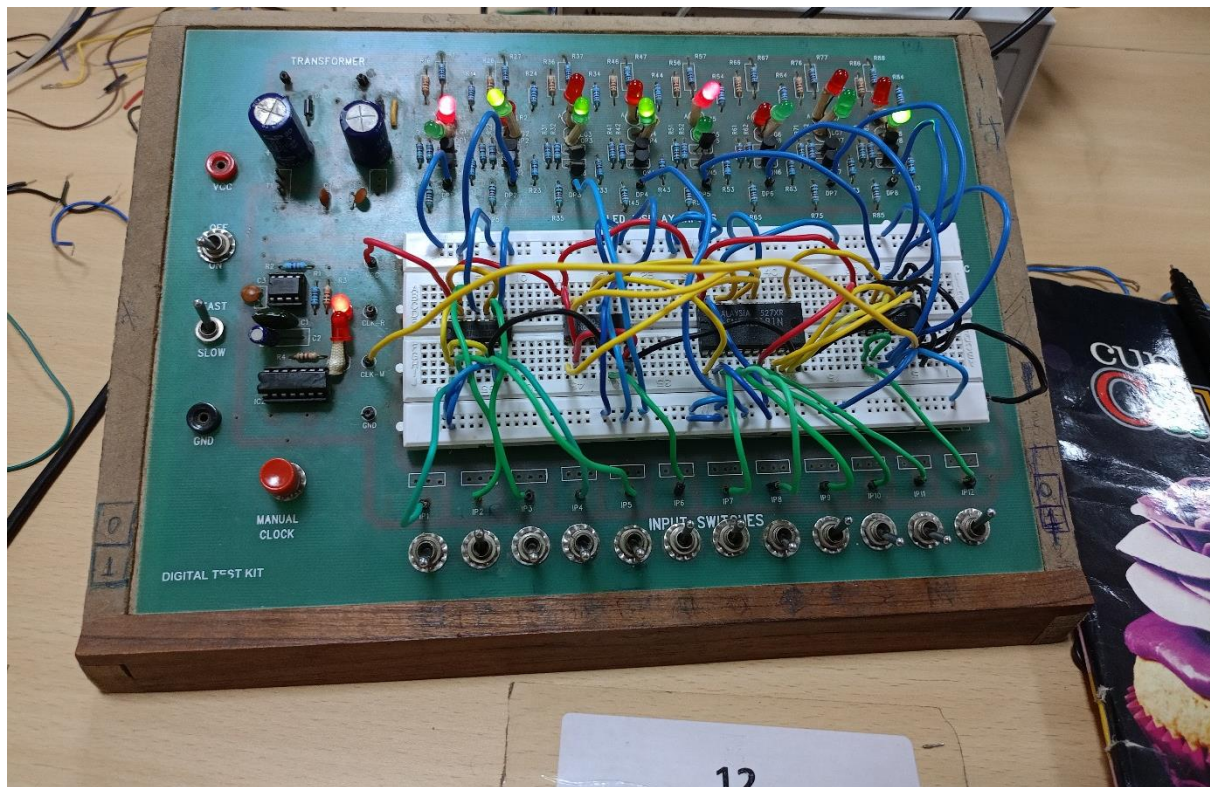
- Digital test kit (w/ breadboard)
- 74LS125 (Tri-State Buffer) IC
- 74LS181 (4-bit ALU) IC
- CD4035 (4-bit Register) IC
- CD4076 (4-bit Register) IC
- Wires

The reference circuit:



Procedure:

- Plug the ICs into the breadboard and power them up.
- Build the circuit as shown in the reference diagram. Use the four horizontal strips on the top and bottom for the four bus lines.
- Ensure the following wire color code:
 - Bus and LED connections with blue wires.
 - Switch inputs to TSB and ALU with green wires.
 - All other connections with yellow wires.
- Before turning on the power, ensure that C' (IP1) is LOW and E_{AR}' (IP12) is HIGH, to avoid bus conflict.
- Provide a 4-bit number (B) via IP2-IP5 onto the bus and load it into the OR via IP6. Apply another number (A) to the bus so that the ALU now has A and B as inputs.
- Perform a suitable ALU operation through select lines IP7-IP10, and note the output on DP5-DP8. Tabulate the inputs applied and the outputs observed for 4 logical functions ($M = IP11 = \text{HIGH}$) and 4 arithmetic functions ($M = \text{LOW}$).
- Now, repeat the operation for any ALU function and note the output. Apply a manual clock pulse to AR, to load the ALU output into the AR.
- Turn C' (IP1) HIGH first, then make E_{AR}' (IP12) LOW, and verify whether the bus output (DP1-DP4) displays the same number as noted before from DP5-DP8 before.
- Verify that the result in AR (DP1-DP4) and the number B in OR give correct result through the ALU output (DP5-DP8).
- Reset the circuit, making C' LOW after making E_{AR}' HIGH again.
- Take 3 4-bit numbers, say A, B and C. Load A into the OR and provide B onto the data bus. Set DP7-DP10 as 1001 (ALU function for A PLUS B).
- Load the ALU output into AR and disable the tri-state buffer ($C' = \text{HIGH}$). Enable AR output ($E_{AR}' = \text{LOW}$), so that the result is on the bus, and load it into OR via IP6.
- Disable AR output and enable the tri-state buffer. Provide C via IP2-IP5.
- The final output at DP5-DP8 should indicate the binary sum $A+B+C$.
- These steps may be repeated for sequential addition of more than 3 numbers.



Conclusion: The bus-based digital system worked correctly.

ALU outputs:

[A = 0111 (7) ; B = 0101 (5)]

ALU Select					Function	OUTPUT			
S ₃	S ₂	S ₁	S ₀	M		Q ₃	Q ₂	Q ₁	Q ₀
0	1	0	1	1	B'	1	0	1	0
0	1	1	1	1	A·B'	0	0	1	0
1	0	0	1	1	(A⊕B)'	1	1	0	1
1	1	1	0	1	A+B	0	1	1	1
0	1	0	0	0	A PLUS (A·B')	1	0	0	1
1	0	0	1	0	A PLUS B	1	1	0	0
1	0	1	0	0	(A+B') PLUS (A·B)	0	1	0	0
1	1	0	0	0	A PLUS A	1	1	1	0

Note that the output for (A+B') PLUS (A·B) did not match the theoretical result, due to overflow. This can be accommodated by considering A and B in their 2's complement form.